

“Hopefully DS will not mess up the quiz, I really want to use my Convex Hull slides.”
— overheard in mess

Tournament Problems

In 1883, Lewis Carroll wrote a newspaper article criticizing how the second-best player was determined in a tennis tournament, and suggested how such a task could be done correctly.

Exercise 1. In a tennis tournament with n players, how many matches are required to find the best player? (Making the unrealistic assumption that a better player always wins.)

How (if in any way) is this different from choosing the maximum element of an array?

Exercise 2. What if we want to determine the best *and* second-best player?

How (if in any way) is this different from choosing the maximum and second maximum elements of an array?

Exercise 3. ♠ What if we want to determine the best, second-best, and third-best players?

How (if in any way) is this different from choosing the maximum, second maximum, and third maximum elements of an array?

Exercise 4. Let’s change the question a bit.

Given (i, j) , we want to find a *mediocre* player, i.e., a player who is not in the top i or the bottom j .

Suggest an $O(n)$ time algorithm for this.¹

Exercise 5. Redo the above problems with F1-style tournaments. We have k tracks.

¹Finding the optimal algorithm for this, taking into consideration the constant, is open.